

note

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1 Affine

(Definition) Dataset

$$\mathcal{D} = \{(x, y)_i\}$$

(Definition) Task

$$\mathcal{T} = (\ell, \mathcal{D}^{\text{train}}, \mathcal{D}^{\text{test}})$$

Supervised learning (Probabilistic View)

$$\phi^* = \arg \max_{\phi} \log p(\phi | \mathcal{D})$$

Meta-supervised learning (Probabilistic View)

$$\phi^* = \arg \max_{\phi} \log p(\phi | \mathcal{D}, \mathcal{D}_{\text{prior}})$$

where $\mathcal{D}_{\text{prior}} = \{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_n\}$.

Meta-supervised learning (Probabilistic View, Approximated)

$$\phi^* = \arg \max_{\phi} \log p(\phi | \mathcal{D}, \theta^*)$$

$$\theta^* = \arg \max_{\theta} \log p(\theta, \mathcal{D}_{\text{prior}})$$

Meta-supervised learning (Mechanistic View)

$$\begin{aligned} \theta^* &= \arg \min_{\theta} \frac{1}{M} \sum_{i=1}^M \ell_i(\phi_i, \mathcal{D}_i^{\text{test}}) \\ \text{s.t.} \quad \phi^* &= f_{\text{Adapt}}(\theta, \mathcal{D}_i^{\text{train}}) \end{aligned}$$

$$\begin{aligned} \theta^* &= \arg \min_{\theta} \frac{1}{M} \sum_{i=1}^M \ell_i(\phi_i^*, \mathcal{D}_i^{\text{test}}) \\ \text{s.t.} \quad \phi_i^* &= f_{\text{Adapt}}(\theta, \mathcal{D}_i^{\text{train}}) \end{aligned}$$

$$\begin{aligned} \phi_i^* &= f_{\theta}(\mathcal{D}_i^{\text{train}}) \\ \phi_i^* &= \arg \min_{\phi_i} \ell_i(\phi_i, \theta, \mathcal{D}_i^{\text{train}}) \end{aligned}$$

$$\begin{aligned} \theta^* &= \arg \min_{\theta} \frac{1}{M} \sum_{i=1}^M \ell_i(\phi_i^*, \mathcal{D}_i^{\text{test}}) \\ \text{s.t.} \quad \phi_i^* &= \arg \min_{\phi_i} \ell_i(\phi_i, \theta, \mathcal{D}_i^{\text{train}}) \end{aligned}$$

$$\phi_i^{k+1} = \phi_i^k - \lambda \nabla_{\phi_i^k} \ell_i(\phi_i^k, \mathcal{D}_i^{\text{train}}), \quad \phi_i^0 = \theta$$

$$\begin{aligned} x_{1:T}^*, u_{1:T}^* &= \arg \min_{x_{1:T} \in \mathcal{X}, u_{1:T} \in \mathcal{U}} \sum_{t=1}^T C(x_t, u_t) \\ \text{s.t.} \quad x_{t+1} &= f(x_t, u_t), \quad x_1 = x_{\text{init}} \\ x_{\min} &\leq x_t \leq x_{\max}, \quad u_{\min} \leq u_t \leq u_{\max} \end{aligned}$$

$$\begin{aligned} \theta^* &= \arg \min_{\theta} \sum_{t=1}^T \|x_t^* - x_t^{\text{ref}}\|_2^2 + \|u_t^* - u_t^{\text{ref}}\|_2^2 \\ \text{s.t.} \quad x_{1:T}^*, u_{1:T}^* &= \arg \min_{x_{1:T} \in \mathcal{X}, u_{1:T} \in \mathcal{U}} \sum_{t=1}^T C_{\theta}(x_t, u_t) \\ \text{s.t.} \quad x_{t+1} &= f_{\theta}(x_t, u_t), \quad x_1 = x_{\text{init}} \\ x_{\min} &\leq x_t \leq x_{\max}, \quad u_{\min} \leq u_t \leq u_{\max} \end{aligned}$$

$$\begin{aligned} \theta^* &= \arg \min_{\theta} \sum_{t=1}^T \|\hat{x}_t - x_t\|_2^2 \\ \text{s.t.} \quad &\hat{x}_t = f_{\theta}(x_{t-1}, u_{t-1}) \end{aligned}$$

$$\begin{aligned} \min_{\theta} \frac{1}{NMT} &\left(\sum_{i=1}^N \sum_{j=1}^M \int_0^T \|x_{ij} - x_i^{\text{ref}}\|_2^2 + \mu_{\text{ctrl}} \|u_{ij}\|_2^2 dt + \mu_{\text{meta}} \|\theta\|_2^2 \right) \\ \text{s.t.} \quad &\dot{x}_{ij} = f(x_{ij}) + B(x_{ij})(u_{ij} + \hat{A}y(x_{ij}; \phi)) \\ &u_{ij} = \bar{\pi}(x_{ij}, x_{ij}^{\text{ref}}, u_{ij}^{\text{ref}}; \kappa) - \hat{A}y(x_{ij}; \phi) \\ &\dot{\hat{A}} = \Gamma B(x)^T \nabla_{x_{ij}} \bar{V}(x_{ij}, x_{ij}^{\text{ref}}; \kappa) y(x; \phi)^T \end{aligned}$$

$$\begin{aligned} \theta &= (\phi, \Gamma, \kappa) \\ \min_{\varphi} \end{aligned}$$

$$\begin{aligned} \dot{x} &= f(x, u, \omega) \\ \dot{x} &= f(x) + g(x)(u + w(x)) \end{aligned}$$

$$\begin{aligned} \|\omega\| &\leq d \\ w(x) &= \theta^T [\omega_1(x), \omega_2(x), \dots, \omega_l(x)] \\ u &= \pi(x) \\ u &= \pi(x, \hat{\theta}) \\ \dot{\hat{\theta}} &= \rho(x) \\ \hat{\theta} &\rightarrow \theta \\ \omega_i(x) \end{aligned}$$

Definition

Dataset

$$\mathcal{D} = \{(x, y)_i\}$$

$$\begin{array}{c} \theta^* \\ \mathcal{D}_i^{\text{train}} \\ \mathcal{D}_i^{\text{test}} \\ \mathcal{D}^{\text{train}} \\ \mathcal{D}^{\text{test}} \end{array}$$